Voice Communications Planning

Part 1. Overview

Terminal learning objective. At the end of this session, you will be able to develop comprehensive voice communications plans during signal operations.

ELO	Objective	Level
8.3.1	Discuss the communication requirements in voice communication planning	C3
8.3.2	Determine communication technologies	C3
8.3.3	Apply voice communication plans	C3
8.3.4	Discuss voice communication resources	C3

References. PAM 6-0102 Signal Operations; PAM 6-00 Command and Control Communications and Cyber Systems (C4S); PAM 3-00 Army Operations (background only).

How to read this handout. Text marked *Manual* comes from the PAMs, with paragraph numbers. Text marked *Supplementary* is teaching material that is not in the manuals; treat it as practice guidance, not doctrine.

Why voice matters. PAM 6-0102 (0134) states that despite computer networks and satellite communication, voice communication remains the bedrock of tactical communication. PAM 6-00 (0205) calls the communications system the voice of command, because it carries the commander's information down to lower commands and units.

Part 2. Voice communication requirements (8.3.1)

2.1 Where requirements come from

Manual: PAM 6-0102 0305. The signal planner obtains and evaluates requirements from:

- the commander's guidance or intent, including the concept of operations;
- staff guidance, because staff officers are the users;
- guidance from higher, subordinate and adjacent headquarters;
- standard operating procedures (SOPs);
- the CEOI and CESI. The CEOI says what to do; the CESI explains how to do it.

The commander directs the means to be provided and the priorities.

2.2 The signal estimate

Manual: PAM 6-0102 0302 to 0303. The signal estimate examines all factors that may affect signal support. METAL is the memory aid: **M**ission, **E**nemy signal capabilities, **T**roops available, **A**rea of operations, **L**ogistical requirements. It is also called the C4S estimate and is revised as the situation changes.

2.3 Voice requirements: what to state

Supplementary. The manuals do not give a worksheet, so use these fields for each voice requirement:

Field	Question
Net or service	Command net, administrative and logistics net, fires, telephone, conference, emergency
Users	Who talks to whom, and how many stations
Mobility	Fixed, on the halt, or on the move
Reach	Distance, terrain between stations, beyond line of sight
Precedence	Who gets the net first when traffic is heavy
Security	Clear or encrypted; what may be said
Availability	Hours, and how quickly it must be restored
Quality	Is intelligible voice enough, or is a telephone-quality call needed

Use Annex A to record them.

2.4 Link requirements to the mission

A requirement is worth stating only if it supports a decision or an action. For each one, ask: which commander's information requirement or task needs this voice link, and what happens if it fails?

Part 3. Voice communication technologies (8.3.2)

3.1 The three categories

Manual: PAM 6-0102 0132 to 0135; PAM 6-00 0402 to 0405. Communications are fixed, mobile (tactical) and special purpose.

Category	What it includes	Voice use
Fixed	BONTEX; PANET; AFP fixed communications system; DBTOCS extends fixed communications to brigades (PAM 6-0102 0133)	Telephone; the Army is shifting to IP telephony on PANET
Tactical	HF, VHF and UHF radios; combat net radio using radio over IP and ad-hoc networks; non-military radios for other agencies (PAM 6-0102 0134)	Radio nets, on the move and on the halt
Special purpose	Commercial video-conferencing, cellular phones, satellite phones (PAM 6-0102 0135)	Routine and administrative traffic and gap fillers, used with prudence and within COMSEC policy

3.2 Fixed system components

Manual: PAM 6-00 0403. Transmission facilities (AFP FCS), switching facilities, cable facilities, terminal facilities (telephones and computers) and support facilities (buildings, power, environmental control).

3.3 Radio bands

Manual: PAM 6-00 glossary and 0514.

Band	Frequency	Typical use
HF/SSB	3 to 30 MHz	Long-distance ground communication; 20 W manpack fielded
VHF/FM	30 to 80 MHz	Short-distance communication; 5 W handheld fielded
UHF/FM	403 to 433 MHz	Internal communications

Supplementary. VHF and UHF depend on line of sight between stations, so hills block them and repeaters on high ground extend them. HF can reach over the horizon, but conditions vary with time of day and season and reception can be noisy.

3.4 Connectivity by command level

Manual: PAM 6-00 0406.

Link	Primary	Secondary	Tertiary or other
Between strategic headquarters	AFP FCS	HF/SSB	Commercial landline; cellular while on the move
Strategic to operational	AFP FCS, with the Army providing drop and insert to major camps	HF/SSB	Commercial landline; mobile if funds permit
Operational to tactical	AFP FCS if technically feasible	HF/SSB with VHF/FM or UHF/FM	Repeaters at strategic areas; commercial landline or mobile as appropriate
Tactical to operating units	AFP FCS if feasible	HF/SSB	VHF/FM, UHF/FM; commercial landline or mobile as appropriate
Unit headquarters to elements	Mobile communications systems	Standardized equipment for interoperability	

3.5 Spectrum, threat and security

- *Spectrum management* is planning, coordinating and managing use of the spectrum under the NTC laws and policies (PAM 6-0102 0218).
- *Electronic warfare* has three parts: support (sense), attack and protect, including frequency hopping and anti-jamming (PAM 6-0102 0223).

- *Vigilance net.* Reserved for emergency and exigent situations and used in joint operations. It is less secure and open to eavesdropping, so use ciphertext and double talk (PAM 6-0102 0316).
- *Compromised frequencies.* Radios may fall into enemy hands, so the plan must include procedures to change frequencies quickly (PAM 6-0102 0311d).

3.6 Choosing a technology

Supplementary. Work through these questions for each link: Distance and terrain? Line of sight available? Does it move? How exposed is it to jamming and interception? What is the weather risk? Which means has a different failure mode as an alternate?

Part 4. Voice communication plans (8.3.3)

4.1 The planning process

Manual: PAM 6-0102 0304 to 0314. The signal planning process runs in conjunction with the MDMP and troop leading procedures.

Step	Paragraph	Content
1. Develop requirements	0305	Sources of requirements
2. Determine available resources	0306 to 0309	Equipment, personnel and funds
3. Develop priorities	0310	Compare requirements with capabilities; rank what can be supported
4. Develop recovery procedures	0311	How to restore service after failure
5. Prepare a tentative plan	0312	Concept with primary, alternate, contingency and emergency plans; allocate resources; site centers; security; interconnection and alternate routing
6. Finalize the plan	0313	Check the plan; reconnaissance and propagation; coverage and gaps
7. Publish plans and orders	0314	Signal plans, the C4S annex, appendices and tabs

4.2 Primary, alternate, contingency, emergency

Manual: PAM 6-0102 0312a. The manual's sample plan shows the idea: the primary means is a Harris handheld VHF tactical radio, the alternate is a Harris manpack tactical radio and the emergency means is a messenger.

Supplementary. Choose each level so it fails differently from the one above it. Two radios on the same repeater are not an alternate.

4.3 Recovery

Manual: PAM 6-0102 0311. Plans must include recovery procedures. This includes changing frequencies when equipment is compromised (0311d) and employment of the cyber response team where network attack is expected (0311c).

4.4 Joint, coalition and disaster operations

Manual: PAM 6-0102 0315 to 0326.

- Joint: vigilance net; lending equipment to another service; embedding Army radio operators in other services, or embedding their operators with our nearest tactical operations center.
- Coalition: each nation keeps a sovereign network, and the plan identifies points of integration.
- Disaster response: the radio net must be different from and separate to the Army's secured operational net. It is treated as unclassified and unsecured, and it connects many entities.

4.5 Net structure

Supplementary. Give each net a purpose, a net control station, a member list, a primary and alternate frequency, and a rule for who may break in. Use the CEOI for call signs and frequencies and the CESI for how to use them.

4.6 Publishing

Manual: PAM 6-0102 0314. Publish the signal plan as the C4S annex with appendices and tabs. Use Annex C of this handout as a working template for the voice appendix.

Part 5. Voice communication resources (8.3.4)

5.1 Equipment, personnel and funds

Manual: PAM 6-0102 0306 to 0309. Keep an up-to-date chart of available C4S equipment, personnel and funds.

- **Equipment (0307):** authorized (TE), on hand, in use or serviceable, unserviceable, reserve pool, and the capability of each item.
- **Personnel (0308):** authorized (TO), actual fill-up, training status, critical shortages, committed or deployed, and reserve.
- **Funds (0309):** installation, operation and maintenance of C4 systems and administrative requirements.

5.2 Maintenance

Manual: PAM 6-0102 0139 to 0142; PAM 6-00 0429 to 0430.

Facility	Tasks
Signal maintenance and training facility (SMTF)	Level 1 to 3 repairs of tactical HF and VHF radios; level 1 to 2 repairs of UHF commercial radios; radio programming; training

Facility	Tasks
Signal mobile maintenance and repair team (SMMART)	Level 1 to 2 repairs of UHF commercial radios; radio programming; refresher training; visits brigades, battalions and companies

5.3 People and training

Manual: PAM 6-00 0447 to 0450, 0525 and the personnel table. Skills are perishable; radio operators of non-signal units need refresher training. Relevant specialties: 22E radio operator/maintainer, 22L radio programmer/technician and 22P electromagnetic spectrum manager. Signal career courses must teach level 1 maintenance of tactical radios.

5.4 Acquisition and complementary systems

Manual: PAM 6-00 0407 and 0514. The Army has acquired the HF/SSB 20 W manpack and the VHF/FM 5 W handheld through the AFP modernization program, with capability enhancement depending on capital outlay funds. Commercial phones and radios may complement tactical systems, and facilities may be leveraged for services, without sacrificing security.

Part 6. Review questions

1. List the sources of voice requirements in PAM 6-0102 0305.
2. Which means is the primary between strategic and operational headquarters, and which is the secondary?
3. Why should a handheld VHF radio net not be the only means for a company in the mountains?
4. What must a plan say about compromised frequencies?
5. Why must a disaster response net be separate from the operational net?
6. Name the three resource areas a planner charts, and give one consideration for each.
7. Which facility repairs level 3 HF radios, and which visits companies?

Part 7. Case scenario: Task Force TALA

Your group is the signal staff of Task Force TALA. You will produce a voice communications plan for a two-phase operation. *All units, places and names are fictional. Where the scenario is silent, state an assumption.*

7.1 Situation

Area. The fictional province of Luntian. **Isla Alon** has a coastal lowland with towns and cellular coverage, and a steep forested mountain range inland. **Isla Bundok** lies 40 km across open water to the east, is hilly, and has no reliable commercial coverage. Typhoon season has started.

Enemy. The Luntian Armed Group (LAG) is in the Isla Alon mountains with cells on Isla Bundok. It uses scanners on clear VHF and cellular traffic, may jam local VHF channels, and may capture radios in ambush.

Mission. Task Force TALA conducts security operations from D-day to D+10 to neutralize the LAG and supports disaster response as directed.

- **Phase 1, D-2 to D+4:** battalions move into the mountains and Isla Bundok.
- **Phase 2, D+5 to D+10:** a typhoon strikes and the task force supports disaster response while continuing operations.

Friendly forces.

Element	Location	Notes
Task Force TALA TOC	Camp Dagat, Isla Alon lowlands	Fixed; connected to PANET and AFP FCS by one microwave link; IP telephones
Forward command post	Mountain foothills, 20 km from the TOC	Mobile, on the halt
1st Battalion	Isla Alon mountains	Dismounted; some companies out of line of sight of the forward CP
2nd Battalion	Isla Bundok	Reaches the TOC only over water
Division headquarters	Outside the area	Reached through PANET and AFP FCS
Navy and local agencies	Coast and provincial offices	Join in Phase 2; use commercial radios and cellular

7.2 Signal assets

Item	Quantity and notes
HF/SSB manpack radios	10
VHF/FM handheld radios	48; 20 encryption-capable, the rest clear voice only
UHF commercial radios	16
Repeater sets	3
Satellite phones	3
Commercial base radios	6, for working with civil agencies
IP telephones at the TOC	8, on PANET
Commercial cellular	Available in the Isla Alon lowlands only
Signal platoon	1 officer and 24 enlisted: 6 radio operator/maintainers, 1 radio programmer, no spectrum manager
Maintenance	SMMART available on request; SMTF at the rear

Item	Quantity and notes
Power	Mains at the TOC; batteries and vehicle generators in the field
Funds	Routine commercial service and consumables only; more must be requested

7.3 Injects

Inject	Event
1 (D+1)	A 1st Battalion company is ambushed and two handheld radios loaded with operational frequencies are captured.
2 (D+2)	A company deep in the mountains loses contact with the forward CP. The terrain blocks the VHF net.
3 (D+3)	Jamming begins on the main VHF battalion command net.
4 (D+5)	The typhoon cuts the microwave link and cellular service. Task Force TALA is ordered to support disaster response.
5 (D+6)	A navy patrol boat and provincial officials ask to join the task force radio net.
6 (D+7)	Handheld batteries are running out, three radios have failed, and operators are exhausted.

7.4 Your tasks

Produce the following as a group:

1. **Condensed signal estimate** using METAL. Maximum two pages.
2. **Voice communications plan** containing:
   - at least eight voice requirements with precedence (8.3.1);
   - the technology for each link, with the reason, tied to terrain, threat and the PAM 6-00 0406 hierarchy (8.3.2);
   - a net diagram and a PACE table for at least six voice services (8.3.3);
   - a procedure for changing frequencies after a compromise (8.3.3);
   - a resource plan covering equipment, personnel, funds, maintenance and training (8.3.4);
   - how the disaster response net is kept separate;
   - your response to each of the six injects.
3. **Brief** of ten minutes to the commander (instructor), then five minutes of questions.

Suggested time. Planning 90 minutes, briefings 15 minutes per group, discussion 15 minutes.

7.5 Evaluation

Criterion	What the evaluator looks for	Objective	Points

Criterion	What the evaluator looks for	Objective	Points
Requirements	Eight or more clear voice requirements linked to the mission, with users, reach, precedence and security	8.3.1	20
Technology selection	Each means fits terrain, threat and hierarchy; reasons stated; limits acknowledged	8.3.2	20
Plan and process	Follows the planning process; net diagram; PACE with independent paths	8.3.3	20
Recovery and security	Frequency compromise, jamming, loss of contact and separate disaster net handled	8.3.3	15
Resources	Equipment, personnel, funds, maintenance and training addressed, with shortfalls and fixes	8.3.4	20
Brief and teamwork	Clear, in time, answers with evidence from the plan	TLO	5
Total			**100**

A group that earns at least 80 points, with no criterion below half its points, has demonstrated the terminal learning objective.

Annex A. Voice requirements worksheet

Supplementary.

No.	Net or service	Users	Mobility	Reach and terrain	Precedence	Security	Availability	Quality
1								
2								
3								
4								

Annex B. PACE table

Supplementary.

Service	Primary	Alternate	Contingency	Emergency	Trigger to change

Annex C. Voice appendix template

Supplementary.

1. Concept of voice support and priorities
2. Net list: purpose, net control, members, frequencies, call signs (by CEOI)
3. Link and technology table with reasons
4. PACE table
5. Frequency compromise and jamming procedures
6. Security measures and the separate disaster net
7. Resources: equipment, personnel, funds, maintenance, training
8. Recovery responsibilities
9. Assumptions

Annex D. Glossary

- **BONTEX:** Bonifacio Telephone Exchange
- **CEOI / CESI:** communication-electronics operating / standing instructions
- **CNR:** combat net radio
- **DBTOCS:** deployable brigade tactical operations center system
- **HF/SSB:** high frequency / single side band
- **PANET:** Philippine Army Network
- **Radio net:** two or more radios on the same frequency for exchanging information
- **ROIP:** radio over IP
- **SMMART:** signal mobile maintenance and repair team
- **SMTF:** signal maintenance and training facility
- **VHF/FM, UHF/FM:** very / ultra high frequency, frequency modulation